Conditional Irrationality of All Projective Stabilizations of Cubic Threefolds: Point-Class Rank under Quantum Wall Crossing

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Abstract

Assume that every smooth projective birational map admits a Włodarczyk completion with marked threshold isomorphisms of the finite cyclic Rees z-differential modules generated by the Gamma point row. We prove that $X \times \mathbf{P}^m$ is irrational for every smooth complex cubic threefold X and every $m \geq 0$.

The endpoint contrast is unconditional. Cai's small quantum-connection matrices give a rank-two block with primitive-sixth formal monodromy. We reconstruct its indicial polynomial directly, and an elementary ${}_2F_3$ Barnes calculation shows that the Gamma point row is nonzero on both primitive-sixth lines. Under multiplication by $\mathbf{P}^m$, quantum Künneth produces $m+1$ copies and preserves detection of the resulting packet, whereas projective space itself has no primitive-sixth packet.

For the global comparison, the Gamma integral structure turns the class of a point into the flat Euler covector which reads ordinary rank. We prove an exact ambient point-column identity at a simple VGIT wall's extremal specialization and exact point-row transport across an ordinary flop on a fixed continuation domain. In a global cobordism, a common-open orbit cylinder gives one endpoint point row, rotation localization kills every intermediate fixed stratum, and Woodward's clutching factorization packages every remaining bubble tree into endpoint tails. The rank-one derived clutching theorem proved here makes each such tail holonomic and tempered. The remaining analytic input is locally finite: at each sign or stability threshold, and on the row-generated nearby-cycle modules at each zero-mode rank change, an isomorphism of the cyclic Rees z-modules must intertwine formal monodromy, carry the marked row, and preserve the stated Stokes, deck, and nearby-cycle data. Under this hypothesis, Rees homogenization makes point-row nonvanishing on a formal-monodromy primary packet birationally invariant. A two-tail rational counterexample shows that separate holonomicity does not determine the threshold map. The incomplete-Gamma and Fourier-boundary countermodels rule out still weaker replacements based only on formal monodromy, pairing, integrality, or localized Fourier support.

Keywords. Gamma class; quantum cohomology; birational wall crossing; VGIT; ordinary flop; Stokes matrix; Fourier–Laplace transform; GKZ system.

1 Introduction

Let Y be a smooth projective complex variety. Iritani's Gamma integral structure associates a flat section $s_Y(E)$ of the quantum connection to a topological K-class E and identifies the flat pairing with the Euler pairing [Iri23]. If $p \in Y$ is a point, then

$$[s_Y(E), s_Y(\mathcal{O}_p)) = \chi(E, \mathcal{O}_p) = \mathrm{rk}(E). \tag{1.1}$$

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The second entry in (1.1) is therefore a distinguished flat covector. We call it the *Gamma point row.*

We study whether the zero or nonzero restriction of this row to a formal monodromy packet survives birational modification. A direct composition of one-wall quantum comparisons does not answer the question: adjacent walls use incompatible Novikov completions, and changing a sectorial lift can add a rank-visible Stokes correction. The main construction instead puts the two birational endpoints in one equivariant cobordism and compares their localized gauged theories before either exceptional completion is taken.

The resulting conditional invariant gives a precise route to stable irrationality for every smooth cubic threefold.

Theorem 1.1 (Conditional cubic stabilization criterion)**.** *Let $X \subset \mathbf{P}^4_{\mathbf{C}}$ be a smooth cubic threefold. Assume that every smooth projective birational map admits a Włodarczyk completion satisfying marked threshold compatibility in the sense of Hypothesis 8.6. Then $X \times \mathbf{P}^m$ is irrational for every $m \geq 0$.*

The $X \times \mathbf{P}^1$ case has a separate dimension-four proof by weak factorization [Rud26]. We keep that argument separate: the present proof introduces a global gauged-localization mechanism which is independent of the stabilization dimension. The case $m = 2$ is the first one not covered by the dimension-four argument.

The conditional conclusion is independent of the classical decomposition of the diagonal and integral Hodge shadows of stable rationality. The companion paper exhibits universally CH_0-trivial cubic threefolds, for which that cycle-theoretic obstruction vanishes, while their first stabilizations remain irrational [Rud26]. The present invariant records irregular quantum-monodromy data instead; it makes no assertion about the Hodge conjecture.

The unconditional endpoint contrast is

$$b(X \times \mathbf{P}^m, \mathcal{P}_6) = 1, \qquad b(\mathbf{P}^{m+3}, \mathcal{P}_6) = 0. \tag{1.2}$$

where an empty packet has value zero. Here b records whether the point row is nonzero on the indicated packet. Section 9 proves (1.2). Cai's cubic connection supplies the primitive-sixth block; we reconstruct its indicial polynomial from his matrices and compute the point coefficient independently from the cubic hypergeometric period. Thus the cubic-specific source dependency is confined to one endpoint lemma.

The birational statement behind the application is more general.

Theorem 1.2 (Conditional point-row primary invariance)**.** *For smooth projective birational complex varieties whose global cobordism satisfies Hypothesis 8.6, the Gamma point row is nonzero on a formal-monodromy primary packet at one endpoint if and only if it is nonzero on the corresponding packet at the other endpoint.*

The conditional proof uses a smooth projective equivariant completion of one Włodarczyk cobordism. A point in the common birational open sweeps out an orbit cylinder. Its equivariant class restricts to the two endpoint point classes and vanishes on every intermediate fixed stratum, so rotation localization removes every wall contribution.

The rest of the proof has two stages. First, Woodward's factorization packages the complete bubble tree in each neutral affine gauge direction into fixed stable-map factors. Theorem 8.4 identifies their fixed stacks, perfect obstruction theories, virtual classes, and evaluations along each affine tail. The moving factors then give a finite-jet Gamma recurrence, so each tail is holonomic and tempered. Second, Hypothesis 8.6 identifies the finite cyclic Rees z-modules across the intervening thresholds while preserving formal monodromy and the marked point row. Nonneutral directions are Laurent-finite, and Rees homogenization makes both endpoint maps intertwine formal monodromy with the same half-Tate shift. The resulting cyclic module determines which primary packets the point row detects.

The remaining hypothesis continues the marked cyclic module, not the complete nonlinear quantum source. Its verification is a marked connection problem and does not follow from the available linear GLSM contour theorem.

The paper also records two local results which motivated the global construction. The first concerns a smooth projective simple VGIT wall $X_- \dashrightarrow X_+$ in the sense of Gu–Yu–Yu, oriented so that $r_+ < r_-$ [GYY25]. Their comparison is a formal, pairing-preserving decomposition

$$\mathrm{QDM}(X_-) \cong \mathrm{QDM}(X_+) \oplus \bigoplus_j \mathrm{QDM}(S)_j \tag{1.3}$$

over an exceptional Laurent completion, where S is the wall quotient. Choose corresponding points $p_\pm$ in the common open set.

Theorem 1.3 (Common-open point column)**.** *At the extremal specialization used in the Gu–Yu–Yu comparison, the ambient coordinate of the point class is exact when $r_+ < r_-$:*

$$\mathrm{pr}_{X_+} \Phi([p_-]) = [p_+]. \tag{1.4}$$

No assertion is made that every wall coordinate of $\Phi([p_-])$ vanishes.

The proof uses the distinguished master-space lift of the common point. Its wall restriction vanishes. Applying the negative-chamber Fourier isomorphism to the lift and to its wall-frequency derivative shows that the derivative is zero in the specialized completed source. Fourier covariance then makes the positive point image horizontal. A regular-singular recursion kills its apparent positive wall-parameter tail.

Our second result is analytic but crepant. Let $Y \dashrightarrow Y'$ be a projective ordinary flop covered by Lee–Lin–Qu–Wang [LLQW16]. Fix one of their analytic-continuation domains in the extremal variable.

Theorem 1.4 (Ordinary-flop point row)**.** *The analytically continued point section of Y equals the intrinsic point section of Y' on the fixed continuation domain. Consequently the point row has the same zero or nonzero restriction to every sectorially realized formal-monodromy packet whose chosen realization is transported by the graph gauge on that branch.*

Pure extremal curves lie in the exceptional locus and cannot meet a point in the common open set. This computes the point column exactly at transverse Novikov degree zero. A divisor pulled back from the common contraction then kills the transverse tail coefficient by coefficient. This is a theorem about one distinguished column; it does not promote the ancestor comparison of [LLQW16] to full descendant invariance.

The discrepant local case contains a further frame comparison. A formal summand, its sectorial lift, and the intrinsic large-radius Gamma section are different objects. Gu–Yu–Yu prove (1.3) in a Laurent/formal coefficient ring, while Shen–Shoemaker identify Gamma asymptotic classes at an extremal specialization [SS25]. Passing from these results to an intrinsic Gamma point row requires a common pairing-preserving sectorial realization. We state the corresponding one-wall consequence only under that explicit hypothesis.

The negative results show why this distinction is necessary. A rank-two incomplete-Gamma connection has a nonzero Stokes shear from a ζ_6-formal line into a rank-visible ambient line. The example admits a flat integral hyperbolic doubling and remains after tensoring with $\mathbf{Q}[N]/(N^3)$. Localized partial Fourier–Laplace transform kills the constant extension which records the shear. In two Fourier variables, the delta module at the common origin is invisible after either localization but transforms back to a constant full-support module. Boundary support in Fourier coordinates therefore points in the opposite direction from the desired rank conclusion.

At the local level, the surviving two-wall statement is one-sided. Complete ray-ordered corrections must have rank-zero *targets*; their sources and scalar coefficients are irrelevant. Under that hypothesis, a finite factorization preserves the point row by elementary linear algebra. A signed punctual coefficient of an oriented boundary complex is a plausible shadow of these factorwise conditions, but it is not equivalent to them in the local formalism. A single alternating coefficient can vanish by cancellation. Under Hypothesis 8.6, the global construction in Section 8 bypasses this missing marked comparison.

Sections 2–4 establish the point-row formalism and the local identities. Sections 5 and 6 give the countermodels, and Section 7 isolates the local composition condition they force. Section 8 proves the Gamma-ratio reduction and conditional Theorem 1.2 by the common-master method. Section 9 proves (1.2) and Theorem 1.1. Section 10 records the source, analytic, and verification boundaries.

2 The point row and three distinct frames

We fix the pairing convention before discussing wall crossing. Let $\mathcal{S}(Y)$ denote a space of flat sections of the quantum connection on a domain where the Gamma framing is defined. Iritani's section associated with $E \in K^0_{\mathrm{top}}(Y)$ has the form

$$s_Y(E) = L_Y(\tau, z) z^{-\mu} z^{c_1(Y)} (2\pi)^{-\dim Y/2} \widehat{\Gamma}_Y (2\pi i)^{\deg/2} \operatorname{ch}(E), \tag{2.1}$$

up to the standard choice of normalization. The flat pairing is

$$[s_1, s_2) := (s_1(\tau, e^{-\pi i} z), s_2(\tau, z))_Y, \qquad [s_Y(E), s_Y(F)) = \chi(E, F). \tag{2.2}$$

See [Iri23, Section 1].

Definition 2.1 (Gamma point row)**.** For $p \in Y$, the Gamma point row is

$$\mathfrak{r}_{Y,p}(v) := [v, s_Y(\mathcal{O}_p)). \tag{2.3}$$

On Gamma-framed K-classes it is ordinary rank:

$$\mathfrak{r}_{Y,p}(s_Y(E)) = \chi(E, \mathcal{O}_p) = \operatorname{rk}(E). \tag{2.4}$$

Putting the point in the first entry instead would introduce the usual dimension sign. We use (2.3) throughout.

Three levels of structure occur in the arguments below.

(i) A *formal packet* is a generalized formal-monodromy or exponential block of the $z = 0$ formal connection.

(ii) A *sectorial lift* is a choice of actual flat subspace with that formal asymptotic expansion on an admissible sector.

(iii) The *intrinsic point section* is the Gamma-normalized flat section fixed by the large-radius fundamental solution.

A positive-z gauge can identify formal quantum D-modules without identifying (iii) across two incompatible Novikov completions. Likewise, two sectorial lifts of the same formal packet can differ by a Stokes shear. The zero or nonzero restriction of (2.3) is therefore not determined by the formal packet alone.

We write $\mathcal{P}_6(Y)$ for the generalized formal-monodromy block with eigenvalue $\zeta_6 = e^{2\pi i/6}$. When $\mathcal{P}_6(Y)$ appears inside a point-row Boolean, it denotes a specified sectorial lift of that block; a different lift is different data. An empty block has Boolean value zero.

Definition 2.2 (Point-row Boolean)**.** For a sectorially realized formal packet $\mathcal{P} \subset \mathcal{S}(Y)$, put

$$b(Y, \mathcal{P}) = \begin{cases} 1, & \mathfrak{r}_{Y,p}|_{\mathcal{P}} \neq 0, \\ 0, & \mathfrak{r}_{Y,p}|_{\mathcal{P}} = 0. \end{cases} \tag{2.5}$$

The paper's exact wall identities either identify the intrinsic point section directly or remain within one specified receiver. Whenever an additional sectorial realization is required, it is stated as a hypothesis.

3 The exact point column at a simple VGIT wall

Let $X_- \dashrightarrow X_+$ be a simple VGIT wall crossing satisfying the hypotheses of [GYY25, Definition 3.2]: the chamber quotients and wall quotient S are smooth and projective, and the normal weights at the wall are ± 1. Assume $r_+ < r_-$ and write $\nu = r_- - r_+$. Theorem 6.2 of [GYY25] gives a pairing-preserving isomorphism

$$\Phi : \mathrm{QDM}(X_-)^{\mathrm{La,red}} \longrightarrow (\tau_+^{\mathrm{red}})^* \,\mathrm{QDM}(X_+)^{\mathrm{La,red}} \oplus \bigoplus_{j=0}^{\nu-1} (\zeta_j^{\mathrm{red}})^* \,\mathrm{QDM}(S)^{\mathrm{La,red}}. \tag{3.1}$$

Its coefficient ring is Laurent in a fractional wall variable and complete in the remaining Novikov and bulk variables. We use only the extremal specialization of (3.1).

Let $p_\pm$ be corresponding points in the common open set. Gu–Yu–Yu's Lemma 3.27 constructs a unique master-space lift a_p with

$$\kappa_+(a_p) = [p_+], \qquad \kappa_-(a_p) = [p_-], \qquad a_p|_{F_0} = 0. \tag{3.2}$$

The last equality follows from the proof of that lemma: the wall restriction is the polynomial determined by the restriction of $[p_+]$ to the positive exceptional locus, which is zero.

Theorem 3.1 (Exact ambient point coordinate)**.** *After setting the master Novikov and bulk variables to zero while retaining the wall variable, the ambient coordinate of (3.1) satisfies*

$$\mathrm{pr}_{X_+} \Phi([p_-]) = [p_+]. \tag{3.3}$$

Proof. Let $q_{\mathrm{w}} = S_{F_0}$ denote the wall variable, let λ be the equivariant wall frequency, and put $D = \kappa_+(\lambda)$. Lemma 5.10 of [GYY25] applies to a_p because its wall restriction is zero. It gives

$$\mathrm{FT}_{X_-}(a_p) = [p_-] \tag{3.4}$$

at the specialization under consideration. Lemma 5.8 gives only

$$\mathrm{FT}_{X_+}(a_p) = [p_+] + O(q_{\mathrm{w}}). \tag{3.5}$$

Apply Lemma 5.10 again to λa_p. Its wall restriction is zero and

$$\kappa_-(\lambda a_p) = \kappa_-(\lambda)[p_-] = 0 \tag{3.6}$$

by degree. The completed source in Proposition 5.2 is finite free, and the negative Fourier map in Theorem 5.5 and Proposition 5.9 is an isomorphism over that completed ring. Base change to the extremal specialization preserves the isomorphism. Equations (3.6) and Lemma 5.10 therefore imply that λa_p is zero in the specialized completed source. In particular,

$$\mathrm{FT}_{X_+}(\lambda a_p) = 0. \tag{3.7}$$

Fourier covariance, Proposition 4.14(2) of [GYY25], together with the quantum-D-module construction of Proposition 4.21, rewrites (3.7) as

$$\left(zq_{\mathrm{w}}\partial_{q_{\mathrm{w}}} + D\star_{\tau_+(q_{\mathrm{w}})} + (q_{\mathrm{w}}\partial_{q_{\mathrm{w}}}\tau_+(q_{\mathrm{w}}))\star_{\tau_+(q_{\mathrm{w}})}\right)\mathrm{FT}_{X_+}(a_p) = 0. \tag{3.8}$$

Here $\tau_+(q_{\mathrm{w}})$ is the specialized receiver coordinate. Write

$$\mathrm{FT}_{X_+}(a_p) = [p_+] + \sum_{k>0} q_{\mathrm{w}}^k v_k, \qquad \tau_+(q_{\mathrm{w}}) = f(q_{\mathrm{w}})\mathbf{1} + \eta(q_{\mathrm{w}}), \tag{3.9}$$

where $f(0) = 0 = \eta(0)$ and $\eta(q_{\mathrm{w}})$ has positive cohomological degree. Lemma 5.8 gives $v_k \in H^*(X_+)[z]$. Every nonconstant stable map in a positive multiple of the wall fibre is contained in the exceptional locus and misses p_+. Thus the positive-wall quantum corrections and $\eta(q_{\mathrm{w}})$ annihilate $[p_+]$.

We prove simultaneously that every v_k and every coefficient of f vanishes. Suppose this is known below order $m > 0$, and write $[q_{\mathrm{w}}^m]f = c_m$. The coefficient of q_{w}^m in (3.8) is

$$(zm\,\mathrm{id} + D\cup -)v_m + mc_m[p_+] = 0. \tag{3.10}$$

If $v_m \neq 0$, the highest power of z in zmv_m cannot be cancelled by either $D \cup v_m$ or the constant term $mc_m[p_+]$, because v_m is polynomial in z. Hence $v_m = 0$, and then $c_m = 0$. Induction kills the complete tail and the possible unit part of the receiver coordinate, proving (3.3). □

Warning 3.2. Theorem 3.1 does not assert that the centre coordinates of $\Phi([p_-])$ vanish. The continuous Fourier map applies stationary phase to the transformed master fundamental solution. Positive master-Novikov terms can survive even though the classical restriction $a_p|_{F_0}$ is zero. The argument proves the ambient coordinate exactly because the wall-frequency derivative dies in the specialized completed source; polynomiality from Lemma 5.8 then controls the possible receiver unit correction.

The following consequence isolates the extra analytic premise required to turn (3.3) into a Gamma-row statement.

Hypothesis 3.3 (One-wall sectorial realization)**.** At a fixed nonzero wall parameter there is a pairing-preserving sectorial realization of (3.1) on one common admissible sector such that:

(a) the ambient formal summand realizes the intrinsic sectorial solution space of X_+;

(b) the wall summands realize the Gamma spans of the wall functors in the Shen–Shoemaker semiorthogonal decomposition;

(c) the realized point row has ambient coordinate (3.3).

Corollary 3.4 (Conditional wall-local rank identity)**.** *Under Hypothesis 3.3, the point row restricts to the point row of X_+ on the ambient packet and vanishes on every wall packet. In particular, for the whole generalized ζ_6 packet in this one receiver,*

$$b(X_-, \mathcal{P}_6(X_-)) = b(X_+, \mathcal{P}_6(X_+)). \tag{3.11}$$

Proof. The ambient assertion is Theorem 3.1 together with the pairing-preserving realization. A wall functor has image supported on the exceptional locus. Pairing its Gamma class with the point class in the common open gives zero Euler characteristic. Shen–Shoemaker's oriented Gamma asymptotics identify these supported classes with the wall blocks on the chosen common sector [SS25]. Hence the point row vanishes there. Restricting the resulting whole-block identity to the generalized ζ_6 packet gives (3.11). $\square$

The hypothesis is not a reformulation of the cited formal theorems. Gu–Yu–Yu work over an exceptional Laurent completion; Shen–Shoemaker's asymptotic theorem is extremal and sectorial. Identifying the intrinsic large-radius point section with the full ambient packet across those frames is additional data.

4 A path-local theorem for ordinary flops

Let $Y \dashrightarrow Y'$ be a projective ordinary flop covered by [LLQW16], and let ℓ, ℓ' be its effective extremal fibre classes. The graph correspondence F identifies the big quantum products and Poincare pairings after analytic continuation

$$q' = q^{-1}, \qquad F(\ell) = -\ell'. \tag{4.1}$$

The continuation is analytic in the extremal variable and formal in the other Novikov variables.

Fix a simply connected nonsingular domain U for the continued extremal variable. We use the hybrid coefficient ring

$$\mathcal{O}(U)\widehat{\otimes}\Lambda_\perp[[\tau]], \tag{4.2}$$

where $\Lambda_\perp$ is the transverse Novikov completion and τ denotes the bulk variables; the coefficient recursion below is taken after extension to $\mathbf{C}((z))$. The *graph gauge* is the horizontal gauge obtained from the Lee–Lin–Qu–Wang comparison of big quantum connections on U, normalized by the classical graph correspondence. Thus it acts on flat columns, not only on quantum products.

Choose corresponding points $p \in Y$ and $p' \in Y'$ outside the exceptional loci. Let A be an ample divisor on the common contraction and write D, D' for its pullbacks. Then $D \cdot \ell = D' \cdot \ell' = 0$, while D is positive on every effective class away from the extremal ray. We complete the transverse Novikov ring by D-degree.

Theorem 4.1 (Point-section preservation)**.** *On U, the graph gauge satisfies*

$$F(s_Y(\mathcal{O}_p)) = s_{Y'}(\mathcal{O}_{p'}). \tag{4.3}$$

Consequently it preserves the Gamma point row.

Proof. At transverse degree zero, every nonconstant stable map has pure extremal class and lies in the exceptional locus. It cannot meet the chosen point. All positive-extremal descendant invariants with the point insertion therefore vanish. The point columns on both sides are their common classical column for every nonsingular value of q: $\widehat{\Gamma}_Y \operatorname{ch}(\mathcal{O}_p)$ is a scalar multiple of the top-degree class $[p]$, on which higher Gamma and Chern-class terms vanish. Analytic continuation of this identically classical extremal column remains classical, and the graph correspondence sends $[p]$ to $[p']$.

Let

$$\delta = F(s_Y(\mathcal{O}_p)) - s_{Y'}(\mathcal{O}_{p'}).$$

It is jointly flat and has zero transverse constant term. Suppose δ_β is a nonzero coefficient of minimal positive D-degree. Take coefficients modulo the extremal ray. By minimality, every positive-transverse convolution term involves a lower D-degree coefficient and vanishes. Horizontality for the Novikov derivation defined by D therefore gives

$$((D \cdot \beta)\,\mathrm{id} + z^{-1}(D \star_q -))\delta_\beta = 0. \tag{4.4}$$

At transverse degree zero, quantum corrections to $D\star_q$ have pure extremal degree. The divisor axiom multiplies them by $D \cdot \ell = 0$. Thus $D\star_q = D \cup -$ in (4.4). Cup product by D is nilpotent and $D \cdot \beta > 0$, so the displayed operator is invertible over $\mathbf{C}((z))$. This contradiction kills the first possible coefficient. Induction proves (4.3). Flatness of the pairing then preserves the row (2.3). □

Corollary 4.2 (Packetwise Boolean)**.** *Let $\mathcal{P} \subset \mathcal{S}(Y)$ be a sectorially realized formal-monodromy packet whose chosen realization is transported by the Lee–Lin–Qu–Wang graph gauge to a sectorially realized packet $\mathcal{P}' \subset \mathcal{S}(Y')$. On the fixed continuation domain,*

$$b(Y, \mathcal{P}) = b(Y', \mathcal{P}'). \tag{4.5}$$

Proof. The graph gauge preserves the quantum connection, grading, first Chern class, and Poincare pairing. It therefore transports the complete $z = 0$ formal type. Theorem 4.1 identifies the covector on the transported sectorial realization, and (4.5) follows. □

The theorem is narrower than a Gamma/Fourier–Mukai comparison for every K-class. Pure extremal curves need not miss the support of a general class, and the support computation in the first paragraph of the proof would fail. Nor does the argument assert phase independence outside the chosen continuation domain.

5 A minimal ambient-target Stokes shear

Set $\alpha = 1/6$ and consider the rank-two meromorphic connection

$$\frac{d}{dz}\begin{pmatrix} y_0 \\ y_1 \end{pmatrix} = \begin{pmatrix} 0 & z^{-2} \\ 0 & z^{-2} + \alpha z^{-1} \end{pmatrix}\begin{pmatrix} y_0 \\ y_1 \end{pmatrix}. \tag{5.1}$$

Proposition 5.1 (Incomplete-Gamma countermodel)**.** *Connection (5.1) has a formal line of monodromy ζ_6 whose sectorial lift undergoes a nonzero Stokes shear into a monodromy-one line. The shear can be equipped with a flat hyperbolic pairing preserving a $\mathbf{Z}[\zeta_6]$-lattice and can be tensored with $\mathbf{Q}[N]/(N^3)$ without disappearing. Hence formal monodromy, pairing, integrality, and a commuting length-three nilpotent tag do not determine the zero or nonzero restriction of a flat covector to the ζ_6 line.*

Proof. Two flat columns are

$$f_0 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad f_\alpha(z) = \begin{pmatrix} \Gamma(1-\alpha, 1/z) \\ z^\alpha e^{-1/z} \end{pmatrix}. \tag{5.2}$$

Indeed,

$$\frac{d}{dz}\Gamma(1-\alpha, 1/z) = z^{\alpha-2}e^{-1/z}, \tag{5.3}$$

and direct differentiation verifies both rows of (5.1). The formal factors are the trivial line and $e^{-1/z}z^\alpha$, whose formal monodromy is $e^{2\pi i\alpha} = \zeta_6$.

Analytic continuation of the incomplete Gamma function adds a nonzero multiple of the complete Gamma value. After rescaling f_α, the Stokes jump has the form

$$f_\alpha^+ = f_\alpha^- + f_0. \tag{5.4}$$

Choose a flat covector r with $r(f_0) = 1$ and $r(f_\alpha^-) = 0$. Then $r(f_\alpha^+) = 1$. Thus the Boolean restriction of r to the sectorial lift of the same formal line changes.

Let $A(z)$ be the matrix in (5.1) and double the system by its dual:

$$B(z) = \operatorname{diag}(A(z), -A(z)^t), \qquad J = \begin{pmatrix} 0 & I_2 \\ I_2 & 0 \end{pmatrix}. \tag{5.5}$$

Then $B^t J + JB = 0$. If the jump on the first summand is $S = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$, the doubled jump $\operatorname{diag}(S, S^{-t})$ preserves J and the obvious $\mathbf{Z}[\zeta_6]$-lattice. Finally tensor with $\mathbf{Q}[N]/(N^3)$ and let the jump act trivially on this factor. It commutes with N and still has (5.4). This proves the assertion. □

The example also occurs in a flat confluence family. Replacing z by z/t gives the z-equation

$$\frac{dY}{dz} = \begin{pmatrix} 0 & tz^{-2} \\ 0 & tz^{-2} + \alpha z^{-1} \end{pmatrix} Y, \tag{5.6}$$

with compatible t-equation inherited from the pullback connection. The two exponential factors collide at $t = 0$, while the nonzero Stokes coefficient persists on the ramified punctured t-line. Formal information at the confluent fibre cannot recover its target orientation.

Remark 5.2. Proposition 5.1 is an analytic no-go theorem, not a geometric counterexample. We do not assert that (5.1) is the quantum connection of a smooth projective wall crossing. It proves only that a geometric proof must use information absent from the listed formal shadows.

6 What localized Fourier–Laplace forgets

The incomplete-Gamma jump has an elementary differential-algebraic shadow. For $g(x) = \Gamma(1 - \alpha, x)$,

$$(\partial_x + 1 + \alpha/x)\partial_x g = 0. \tag{6.1}$$

Differentiation forgets the constant solution, and that constant is the additive datum in the continuation formula.

Let $D_t = \mathbf{C}\langle t, \partial_t\rangle$. Under algebraic Fourier transform,

$$\mathbf{C}[t] = D_t/D_t\partial_t \quad \longmapsto \quad \delta_0 = D_\tau/D_\tau\tau. \tag{6.2}$$

Localizing at τ kills δ_0. Localized partial Fourier–Laplace therefore cannot recover the extension constant in (6.1). The four-term GKZ/Gauss–Manin comparison in [RSSW21] uses the same nonconservativity.

Proposition 6.1 (Punctual Fourier corner)**.** *Let*

$$\delta_{00} = D_{\tau_1,\tau_2}/D_{\tau_1,\tau_2}(\tau_1, \tau_2).$$

Localization at either τ_i kills δ_{00}, whereas

$$\mathrm{FL}^{-1}(\delta_{00}) = \mathcal{O}_{\mathbf{A}^2} \tag{6.3}$$

up to shift and sign conventions. Thus an object supported at the intersection of two deleted Fourier boundaries can have generic-rank-one inverse transform.

Proof. Use the Weyl-algebra Fourier automorphism $x_i \mapsto -\partial_{\tau_i}$ and $\partial_{x_i} \mapsto \tau_i$. It exchanges $D_x/D_x(\partial_{x_1}, \partial_{x_2})$ with δ_{00}. Since τ_i acts locally nilpotently on the latter, inverting either coordinate kills it. The inverse transform is the constant module, which has generic rank one. $\square$

Tensoring Proposition 6.1 with the ζ_6 rank-one connection, its hyperbolic dual, and $\mathbf{Q}[N]/(N^3)$ preserves every conclusion. Bare double-boundary support is therefore not a rank-zero condition.

Nor does marking one punctual vector make it unique. Let C be any finite-dimensional coefficient space and put $\mathcal{B} = \delta_{00} \otimes C$. For $G \in \mathrm{GL}(C)$, two receivers can differ by G while having the same support and characteristic cycle. After hyperbolic doubling, $G \oplus G^{-t}$ preserves a perfect pairing. It can still change a chosen point row. Consequently, a claim that the punctual quotient is generated by one marked line is additional geometry. Enlarging the discarded subspace to the kernel of the point row makes the quotient one-dimensional only tautologically: invariance of that kernel is the theorem one seeks.

The unlocalized boundary map

$$\mathrm{FL}(M) \longrightarrow j_* j^* \, \mathrm{FL}(M), \qquad j : \mathbf{G}_m \hookrightarrow \mathbf{A}^1, \tag{6.4}$$

retains the missing object. In a two-coordinate problem one must also retain the orientation, duality, and $! \to *$ or can / var maps of the complete Cech boundary square. Proposition 6.1 shows that retaining the object is necessary, not that its support alone proves the desired vanishing.

7 The two-wall criterion

Let Y be an intermediate smooth projective variety, $D \subset Y$ the union of the two incident wall boundaries, and $p \in U := Y \setminus D$. Suppose a common analytic receiver contains the two sectorial realizations under comparison, and let T be their transition. Write $\mathcal{S}_D(Y)$ for a span of Gamma sections represented by classes supported on D. The point row annihilates this span:

$$\mathfrak{r}_{Y,p}(v) = 0 \qquad (v \in \mathcal{S}_D(Y)). \tag{7.1}$$

Hypothesis 7.1 (Rank-zero-target condition)**.** On the packet under consideration, the relative transition admits a canonical ray-ordered factorization

$$T = \prod_{a=1}^{m} (1 + v_a \otimes \lambda_a) \tag{7.2}$$

whose complete residue targets satisfy

$$\mathfrak{r}_{Y,p}(v_a) = 0 \qquad (1 \le a \le m). \tag{7.3}$$

The word *complete* is essential. Refining a residue block into coordinate terms can create nonzero ranks which cancel only after the full block is assembled. The boundary-support condition $v_a \in \mathcal{S}_D(Y)$ is a geometric sufficient condition for (7.3), by (7.1), but it is not part of the abstract hypothesis.

Theorem 7.2 (One-row assembly)**.** *Under Hypothesis 7.1,*

$$\mathfrak{r}_{Y,p} T = \mathfrak{r}_{Y,p}. \tag{7.4}$$

Consequently, any finite factorization whose ordinary-flop steps satisfy Theorem 4.1 and whose discrepant adjacent-wall transitions satisfy Hypothesis 7.1 preserves the point-row Boolean on the transported packet.

Proof. Equation (7.3) gives

$$\mathfrak{r}_{Y,p}(1 + v_a \otimes \lambda_a) = \mathfrak{r}_{Y,p}$$

for every factor in (7.2). Multiplying the identities proves (7.4). The factorization statement follows by induction, using Theorem 4.1 at crepant steps. □

A stronger geometric condition is one-sided kernel support. If the cone of the transition kernel relative to the diagonal has support

$$\operatorname{Supp}(K_{\mathrm{rel}}) \subset Y \times D, \tag{7.5}$$

with D in the output factor, and if its induced action satisfies $\operatorname{im}(T - \mathrm{id}) \subset \mathcal{S}_D(Y)$, then (7.1) gives $\mathfrak{r}_{Y,p}T = \mathfrak{r}_{Y,p}$ directly. This conclusion does not require, and does not imply, that every factor in an independently chosen factorization of T has supported target. Support in $D \times Y$ instead constrains the source and would not imply (7.4).

There is a tempting singular shadow of the factorwise condition. Let B_{corner} be a complete oriented correction left by the two localizations after subtracting the common point contribution, and let its signed punctual multiplicity be the alternating δ_{00} coefficient

$$\mu_{00}(B_{\mathrm{corner}}).$$

Fourier transform exchanges a punctual coefficient with a zero-section multiplicity, hence with generic output rank in the elementary model of Proposition 6.1. One might therefore hope for

$$\mu_{00}(B_{\mathrm{corner}}) = 0 \tag{7.6}$$

as a numerical test. This implication is *not* proved here. A single alternating multiplicity can vanish by cancellation even when individual ray targets have nonzero point row, and no exact point-row marking on B_{corner} has yet been constructed. Proposition 6.1 also explains why merely knowing that the boundary object is punctual proves nothing.

The failure of a global-to-factorwise inference already occurs in dimension two. Let $r = e_1^*$ on $V = \mathbf{C}^2$, and put $A = e_1 \otimes e_2^*$. Then $A^2 = 0$ and

$$\mathrm{id} = (\mathrm{id} + A)(\mathrm{id} - A), \tag{7.7}$$

although both elementary factors have target e_1 and $r(e_1) = 1$. Thus even a vanishing total correction does not force the targets in a chosen ray factorization to lie in $\ker r$.

The missing comparison can be stated blockwise: for each complete ray block one would need an unlocalized boundary object B_a and an identity

$$\mathfrak{r}_{Y,p}(v_a) = \mu_{00}(B_a). \tag{7.8}$$

with enough concentration or effectivity to prevent cancellation inside the block. Equation (7.8) and $\mu_{00}(B_a) = 0$ for every a would imply Hypothesis 7.1; the single global equation (7.6) does not.

Remark 7.3 (What remains to be proved). Neither (7.2), the action condition following (7.5), nor the blockwise comparison (7.8) is deduced here for a general pair of discrepant walls. Existing one-wall theorems determine formal decompositions and, in special toric settings, Gamma/window comparisons. The missing statement is directed and two-wall: it must retain which side of every vanishing cycle is the output and must compare that orientation with the Gamma point row.

8 A common master and birational transport

The obstruction in Sections 5–7 comes from comparing two local receivers at an intermediate chamber. We now avoid that comparison. A single equivariant cobordism carries both endpoint quantum Kirwan maps, and virtual localization compares their point rows before either endpoint Novikov completion is taken.

8.1 The common point row

Let W be a smooth projective variety with a $\mathbf{G}_m$-action and let Y_- and Y_+ be smooth free GIT quotients for two extreme linearizations. We assume the stable-equals-semistable and properness hypotheses under which Woodward's localized gauged theory is defined. In particular, degree by degree, the polarization master stack is a proper Deligne–Mumford stack with its stated perfect obstruction theory. We work coefficientwise at sufficiently large area. Write $\kappa_\pm$ for the corresponding quantum Kirwan maps, write $\mathrm{Kir}_\pm : H^*_{\mathbf{G}_m}(W) \to H^*(Y_\pm)$ for the ordinary cohomological Kirwan maps, and write $\tau_{Y_\pm,-}$ for Woodward's localized graph potentials. The derivatives

$$A_\pm = D\tau_{Y_\pm,-}\, D\kappa_\pm \tag{8.1}$$

are the endpoint localized gauged maps; this is the localized adiabatic identity [Woo18, Equation (68)].

Suppose that a class $a_p \in H^*_{\mathbf{G}_m}(W)$ has the following restrictions:

$$\mathrm{Kir}_\pm(a_p) = [p_\pm], \qquad a_p|_F = 0 \tag{8.2}$$

for every fixed component F occurring at an intermediate polarization. Here p_- and p_+ lie in the common birational open set.

Proposition 8.1 (Support collapse)**.** *At every fixed equivariant degree, the point-marked virtual Kalkman formula has no intermediate-polarization contribution. Hence its two endpoint coefficients agree in the extended Novikov coefficient-function space. After Rees homogenization and in any common realization of the two endpoint series, this reads*

$$\mathfrak{r}_{Y_-,p_-} A_- = \mathfrak{r}_{Y_+,p_+} A_+. \tag{8.3}$$

The identity is coefficientwise in all remaining Novikov and bulk variables; the proposition does not construct the common realization.

Proof. Retain rotation of the parametrized graph curve, with equivariant parameter $\hbar$. At each ordinary input marking insert the normalized equivariant point class of $0 \in \mathbf{P}^1$. Rotation localization then puts every input on the zero-side bubble tree. Next choose equivariant divisor classes $D_1, \ldots, D_s$ spanning the full equivariant numerical dual of the allowed total degrees; their endpoint Kirwan restrictions in particular span the endpoint numerical divisor spaces. Insert Woodward's Liouville classes with parameters t_j. On a graph fixed locus their contribution is

$$\exp\left(\sum_j t_j D_j\right) \prod_j x_j^{D_j \cdot \widetilde{d}_+}, \qquad x_j = \exp(t_j \hbar), \tag{8.4}$$

where $\widetilde{d}_+$ is the total infinity-side equivariant degree: the bubble degree together with the affine cocharacter degree of the gauged principal component [Woo18, Corollary 9.10(c)].

Extracting the trivial character in all x_j forces $\widetilde{d}_+ = 0$. The infinity factor is therefore the unstable identity: positivity on the effective infinity-side semigroup leaves no nonconstant degree-zero component. The zero factor is exactly the localized graph potential in (8.1). Using the full equivariant divisor lattice is important: endpoint divisor spaces alone need not separate an affine cocharacter degree.

For each fixed degree and sufficiently large area, every nonendpoint polarization-fixed graph has its principal component in a fixed quotient. Moreover every marking and node lands in the corresponding fixed locus [GW15, Proposition 3.15(c), Lemma 3.17 and Proposition 3.18]. Its insertion therefore factors through one of the restrictions in (8.2) and vanishes before division by the virtual normal Euler class. Only the two endpoints survive the virtual Kalkman identity. Their signs and node factors agree with Woodward's localized adiabatic identity. At either endpoint the distinguished equivariant insertion restricts through $\mathrm{Kir}_\pm$, hence becomes $[p_\pm]$ by (8.2); the quantum Kirwan derivatives remain inside $A_\pm$. This gives (8.3). Polarizing in the ordinary inputs gives the entire row rather than only the scalar potential. □

For a birational map between smooth projective varieties, the required class exists globally. Włodarczyk's smooth birational cobordism admits a smooth projective equivariant completion whose source and sink quotients are the two endpoints [Wło00, Proposition 2(B')]. Choose p in the common birational open supplied as a trivialized cylinder by the construction. There the cobordism is a trivial orbit cylinder. The equivariant Poincaré dual of the closure of that orbit restricts to the two point classes and misses every intermediate fixed component. It is the class a_p in (8.2). This assertion concerns the ordinary Kirwan restrictions only; no vanishing of positive-degree corrections to the quantum Kirwan maps is used.

There is a terminology point in the source. Włodarczyk calls the open cobordism *projective* when it is quasiprojective. We use the resolved proper equivariant completion constructed in the proof of Proposition 2(B'); the resolution is an isomorphism over the source, sink, and common-open cylinder. Thus the extreme quotients and the class a_p are unchanged.

8.2 Neutral tails and threshold comparison

The two maps in (8.1) initially belong to opposite Novikov completions. The comparison needed for the point-row Boolean is smaller than a common field for the complete source. Fix a finite ample-energy and bulk Artin quotient of the equivariant Novikov ring. For a primitive affine gauge direction δ, all ordinary curve degrees and fixed graph types are then finite; only the integer coordinate along δ can remain unbounded. We first control those unbounded tails.

The argument separates the fixed geometry from the remaining analytic comparison. A rank-one derived-intersection theorem makes the fixed clutching data constant along each affine tail; the moving normal weights then give a Gamma recurrence. Marked threshold compatibility subsequently joins the resulting cyclic z-modules across the finitely many changes of sign, stability, and zero-mode rank.

Call the direction neutral when

$$c_1^{\mathbf{G}_m}(TW) \cdot \delta = 0. \tag{8.5}$$

We compare formal monodromy after the Rees substitution

$$Q^d \longmapsto z^{c_1^{\mathbf{G}_m}(TW)\cdot d} Q^d. \tag{8.6}$$

Proposition 8.2 (Gamma-ratio reduction)**.** *At any finite Artin level, every individual rotation-fixed graph coefficient in a neutral direction is a finite nilpotent derivative of a vector-valued balanced Gamma-ratio kernel. Its signed slope is* $c_1^{\mathbf{G}_m}(TW)\cdot\delta$*. In a nonneutral direction, every homogeneous coefficient is Laurent-finite.*

Proof. Woodward's classification of the rotation-fixed graph locus is exhaustive for sufficiently large area [Woo18, Corollary 9.10]. His virtual-normal splitting gives

$$\mathrm{Eul}(N_\pm) = \mathrm{Eul}((R\pi_*\,\mathrm{ev}^*\,T(W/\mathbf{G}_m))_{\mathrm{mov}})\,(\mp\zeta)(\mp\zeta-\psi). \tag{8.7}$$

The last two factors smooth the node and move the attaching point. They are independent of the affine gauge degree, so gluing neither removes one end of the degree lattice nor creates a degree-dependent pole. Formula (8.7) applies when a nontrivial attached component is present; in the irreducible unstable type those factors are absent and only the moving index remains [Woo18, Example 9.15].

Pass to a finite cover which clears stabilizer denominators and apply the splitting principle. A virtual line in the moving index of the principal weighted component has nilpotent Chern root α_a and degree

$$n_a(k) = h_a k + s_a. \tag{8.8}$$

For every integer n, its inverse index Euler factor is the single meromorphic expression

$$\frac{1}{\prod_{m=0}^{n}(\alpha_a+m\zeta)} = \zeta^{-n-1}\frac{\Gamma(\alpha_a/\zeta)}{\Gamma(\alpha_a/\zeta+n+1)}. \tag{8.9}$$

For $n<0$, the left side denotes the Euler class of H^1; for $n\geq 0$, it denotes the inverse Euler class of H^0. Thus the two degree rays are not assigned unrelated Gamma products.

The signed sum of the moving slopes is

$$\sum_a \epsilon_a h_a = c_1^{\mathbf{G}_m}(TW)\cdot\delta. \tag{8.10}$$

The gauge Lie algebra has slope zero. Moving modes on attached fixed bubbles have fixed rank at the chosen ordinary degree. Their Euler factors are finite products of linear functions of k, possibly inverted. Each inverse factor is an adjacent Gamma ratio

$$(hk+a)^{-1} = \frac{\Gamma(hk+a)}{\Gamma(hk+a+1)}, \tag{8.11}$$

so it has zero signed slope and does not alter (8.10). Equivariant inputs are polynomial in k; Chern roots and psi classes are nilpotent at Artin level; Liouville classes supply the Fourier exponential.

Under (8.5), equations (8.7)–(8.11) therefore give a balanced Gamma-ratio kernel coefficientwise. Polynomial equivariant insertions are Fourier derivatives, while nilpotent Chern-root and bulk shifts give only finitely many parameter derivatives. For comparison, one moving-character residue has the exact normalization

$$\operatorname*{Res}_{h\sigma+\alpha=-m}\Gamma(h\sigma+\alpha)\,d\sigma = \frac{(-1)^m}{h\,m!}. \tag{8.12}$$

The factorial is the moving-mode Euler factor, $1/h$ is the one-character Jacobian factor, and the sign is the complex orientation. We use (8.12) only as the normalization of a single simple pole. Coincident or higher poles require derivatives and the full localization expression, including graph automorphism factors; no product-of-simple-residues formula is asserted here.

If (8.5) fails, the virtual-dimension equation fixes k in each homogeneous coefficient, hence the coefficient is Laurent-finite. This proves the last assertion. □

Remark 8.3 (Exact analytic boundary). Aleshkin–Liu prove the relevant contour statement for a finite-dimensional linear abelian GLSM with adjacent secondary-fan chambers, a circuit, Calabi–Yau charges, and a grade-restriction window [AL23, Definition 5.18 and Theorem 5.21]. Proposition 8.2 does not construct those data for a nonlinear virtual fixed graph. Their theorem therefore supplies neither the derived-intersection argument below nor the marked threshold comparisons.

González–Woodward's crepant wall-crossing theorem gives a weaker distributional statement: neutral Picard shifts produce sums of derivatives of delta distributions [GW15, Remark 1.18(d), Remark 4.6, and equation (47)], and the authors explicitly do not deduce analytic continuation without an independent convergence input [GW15, Remark 4.7]. This does not replace Hypothesis 8.6. Proposition 6.1 exhibits the underlying obstruction: information supported at a Fourier boundary can reappear with full support after inverse transform, so almost-everywhere equality does not identify the marked formal packet.

Theorem 8.4 (Rank-one tailwise derived identification). *For every finite Artin level, ordinary bubble degree, and rotation-fixed graph type, after splitting the affine degrees into stabilizer congruence classes and removing the finite sign, stability, and zero-mode thresholds, there are isomorphisms of the fixed clutching stacks and their universal domains which identify consecutive degrees in each remaining tail. These isomorphisms identify the evaluation maps and the fixed parts of the relative perfect obstruction theories as morphisms to the corresponding cotangent complexes, hence also their virtual classes. They respect stack automorphisms, gluing over the inertia quotient, and reduction between Artin levels. Stable degree-zero attached bubbles are retained as fixed stable-map factors; the irreducible unstable type is treated by its separate convention.*

Proof. Fix the generic semistable locus and the two endpoint fixed components specified by the graph type. Write W_F^a for the corresponding Białynicki–Birula stratum: its points lie in that generic locus and satisfy $\lim_{t\to 0} t^a x \in F$. Evaluation at 1 identifies fixed sections on one parametrized affine chart with W_F^a: the section determined by x is $z \mapsto z^a x$, and it extends across the origin with limit in F exactly when $x \in W_F^a$. A principal clutching section has two such extensions with the same generic value. Let $\mathfrak{S}_{a_-,a_+}$ denote the derived fixed section stack for the chosen clutching bundles, stabilizer lift, and graph component. Evaluation at 1 is a functorial equivalence on derived bases

$$\mathfrak{S}_{a_-,a_+} \simeq \mathfrak{Z} := [W_{F_-}^{a_-}/\mathbf{G}_m] \overset{\mathbf{R}}{\underset{[W/\mathbf{G}_m]}{\times}} [W_{F_+}^{a_+}/\mathbf{G}_m].$$

Here is an affine verification of the derived assertion. On a $\mathbf{G}_m$-invariant Sumihiro chart $\operatorname{Spec} A$, write $A = \bigoplus_w A_w$. For a derived test algebra R, an equivariant map from the parametrized affine chart is a graded morphism $A \to R[z]$; after evaluation at 1, a homogeneous element of weight w has the unique extension $f \mapsto f(x)z^{aw}$. Regularity at the origin says exactly that the coefficients with $aw < 0$ vanish. Thus the functor factors through the derived base change of the attractor algebra $A/(A_w : aw < 0)$, with the component and semistability open imposed. This calculation is valid for simplicial commutative test algebras degreewise, commutes with base change, and glues over the equivariant affine cover. After quotienting by change of trivialization it proves the one-chart equivalence with $[W_F^a/\mathbf{G}_m]$, including stabilizers and nilpotent derived families. Derived descent over the two-chart cover then gives the displayed homotopy fibre product. Its inverse equips a family x with the prescribed two clutching bundles and the sections $z \mapsto z^{a_\pm} x$. The chosen strata impose the generic stability and graph-component conditions. The attached stable-map factors in Woodward's equations (54)–(57) are added by derived fibre products over the two inertia evaluations.

For $\mathbf{G}_m$, replacing a nonzero endpoint exponent by another integer of the same sign leaves the relevant stratum and limit map unchanged. Passing to a fixed stabilizer congruence class also fixes the inertia label. Thus the displayed derived intersection, its universal domain and evaluations, and its derived fibre products with the fixed ordinary stable-map factors are identical throughout one tail. The universal gauged maps $z \mapsto z^a x$ themselves are not identified when a changes; it remains to compare the fixed deformation complexes.

All tangent and cotangent complexes in the next calculation are relative to the fixed moduli of marked domains, denoted $\mathfrak{M}$. The tangent complex of the displayed derived intersection is

$$\left[T(W^{a_-}_{F_-}/\mathbf{G}_m) \oplus T(W^{a_+}_{F_+}/\mathbf{G}_m) \longrightarrow T(W/\mathbf{G}_m)\right].$$

It retains the degree-one term when the two attracting loci meet nontransversely. Equivalently, trivialize the pulled-back tangent complex on the open orbit by evaluation at 1. Taking invariants in the two-chart Čech resolution gives

$$\left[F^0_{\leq 0}V \oplus F^\infty_{\leq 0}V \longrightarrow V\right],$$

where the two terms are the extension subbundles at the endpoints. They are the tangent bundles of the two attracting loci. Thus this invariant Čech complex agrees with the preceding tangent complex without choosing an equivariant splitting. More precisely, in perfect complexes on $\mathfrak{Z}$, functoriality of restriction to the two charts gives

$$(R\pi_* u_a^* T(W/\mathbf{G}_m))^{\mathrm{rot}} \simeq \left[F^0_{\leq 0}V \oplus F^\infty_{\leq 0}V \longrightarrow V\right] \simeq T_{\mathfrak{Z}/\mathfrak{M}}.$$

The composite is the differential of evaluation at 1: on the two-chart cover, both it and the tangent map of the derived fibre product are induced by the same restriction morphisms to the open orbit. This identifies the complex together with its canonical map to the tangent complex, not only its class in K-theory. Along a tail the signs of all endpoint weights are constant, and the zero-weight changes were removed as thresholds. The two extension subbundles and the restriction morphism are therefore constant, even though the universal gauged map varies with a.

The dual of the fixed perfect obstruction theory is the derived pushforward of $T(W/\mathbf{G}_m)$ [Woo18, Definition 7.13]; it is the truncation of the intrinsic tangent complex of this derived section stack. Woodward's cutting-edge compatibility identifies the obstruction theory of the glued fixed graph with the corresponding derived fibre-product triangle [Woo18, Proposition 7.14]. Since the comparison above identifies its invariant Čech model as a perfect complex and the differential displayed above, duality identifies the fixed obstruction-theory morphisms to $L_{t_0\mathfrak{Z}/\mathfrak{M}}$, hence the virtual classes. The cotangent triangle for the derived fibre products over inertia then glues this identification to the unchanged perfect obstruction theories of the fixed ordinary stable-map factors.

On a k-fold orbifold cover the same affine calculation is made in $R[z^{1/k}]$ with its residual μ_k-grading. The stabilizer congruence fixes that grading and the twisted-sector evaluation. Rigidifying the inertia receiver quotients both sides by the same residual gerbe, so the derived equivalence descends. Because it is an equivalence of stacks, stabilizer and graph-automorphism groups, and hence their localization multiplicities, are preserved. The parametrized principal component has no domain automorphisms; the fixed bubble factors retain theirs. Every construction above is by base change and therefore commutes with Artin reduction. The irreducible unstable type is the corresponding localized endpoint-chart calculation with no bubble, node, or attaching factor. This proves every assertion. □

Proposition 8.5 (Clutching-tail holonomicity)**.** *Every point-marked endpoint tail is a finite nilpotent derivative of one Gamma-ratio recurrence with affine arguments. Its generating series is holonomic. In a neutral direction its coefficients have polynomial-logarithmic growth after one exponential rescaling, so the tail has tempered radial boundary values.*

Proof. Corollary 9.10 and equations (54), (56), and (57) of [Woo18] express a rotation-fixed component as a fibre product of two clutching factors over the inertia quotient. Each factor already contains the complete ordinary stable-map space attached at that end. Thus arbitrary bubble trees occur in fixed proper stable-map factors once their ordinary degrees and markings are fixed; no induction over bubble-tree edges is required. Theorem 8.4 supplies the degree-shift identification of those fixed factors, their obstruction theories, and their evaluations. This is stronger than the normal-complex identity in equations (58)–(59) of [Woo18].

For $G = \mathbf{G}_m$, an affine clutching cocharacter is an integer. On one of the identified tails, apply the splitting principle to the moving part of the normal complex. A character root of weight h has degree $hk + s$, so the complementary moving weights give (8.9), with nilpotent Chern roots on the fixed coefficient stack. Stable attached components, including degree-zero bubbles stabilized by markings and nodes, retain the node and attaching factors in (8.7). Only the irreducible unstable type has no such factors. They are adjacent Gamma ratios and do not change the signed slope.

Only finitely many parameter derivatives are therefore needed. Gamma arguments do not cross a pole inside a tail, since every zero-mode crossing was removed as a threshold. Clearing recurrence denominators is therefore legitimate there. Gamma recurrence gives a rational finite-state recurrence in k, hence a holonomic generating series. Under (8.5), Stirling's formula cancels the $k \log k$ terms in the signed product; the remaining exponential is removed by one rescaling and the coefficients grow by a power of k and $\log k$. This gives tempered Fourier growth. Finite threshold terms are Laurent polynomials and do not alter the conclusion. □

Tailwise holonomicity does not identify the two selected solutions. The bilateral sequence

$$c_k = \begin{cases} 2^k, & k \geq 0, \\ 3^{-k}, & k < 0 \end{cases}$$

has oriented germs

$$F_0(x) = \frac{1}{1 - 2x}, \qquad F_\infty(x) = \frac{3}{x - 3}.$$

Both tails are first-order hypergeometric and tempered on their natural boundary circles. Away from the single threshold, the bilateral sequence is annihilated by $(E - 2)(E - 1/3)$; multiplying by a polynomial in the index absorbs the finite defect. Nevertheless the two displayed rational functions are not branches of one meromorphic function. Thus neither a polynomial recurrence nor tailwise holonomicity determines the finite connection data across a threshold.

In the next hypothesis, a *finite dual cyclic Rees z-differential module* means the Rees differential module corresponding, under the sectorial irregular Riemann–Hilbert correspondence, to the smallest finite Stokes-filtered local subsystem which contains the marked horizontal row and is stable under normalized formal monodromy and the ramified deck action. It inherits the flat connection; it is not generated by applying the connection to a horizontal section. Its distinguished formal-monodromy Krylov subspace is $C_T(r)$ from Lemma 8.7.

Hypothesis 8.6 (Marked threshold compatibility)**.** At every finite Artin level and fixed ordinary degree, choose a common ramified admissible z-sector. At a sign or stability threshold, let K_j^- and K_j^+ be the finite dual cyclic Rees z-differential modules of the two adjacent tails. At a zero-mode rank change, write $M_j^\pm$ for those adjacent modules, choose a transverse parameter t, and use the ramified Stokes-filtered nearby- and vanishing-cycle functors in the standard triangle

$$i^*M \longrightarrow \psi_t M \xrightarrow{\text{can}} \phi_t M \xrightarrow{+1}, \qquad \text{var} : \phi_t M \longrightarrow \psi_t M.$$

For either adjacent module M, let $V_t(M) \subset \psi_t M$ be the smallest Stokes-filtered Rees submodule that contains the image of var : $\phi_t M \to \psi_t M$ and is stable under formal monodromy and the deck action. Put

$$\psi_t^{\text{red}} M := \psi_t M / V_t(M).$$

Thus reduced nearby cycles are defined by the nearby/vanishing-cycle triangle, not by a choice of a complementary summand. In this case let $K_j^\pm$ be the smallest subobjects of $\psi_t^{\text{red}} M_j^\pm$ containing the specialized marked rows and stable under the induced monodromy, Stokes filtration, and deck action. The specialization maps from the adjacent tail modules to $K_j^\pm$ are required to preserve and reflect nonvanishing of every primary projection for the induced formal monodromy. Denote the resulting rows by $r_j^\pm$ and the half-Tate-normalized formal monodromies by $T_j^\pm$. There is an invertible comparison

$$\Phi_j : K_j^- \xrightarrow{\sim} K_j^+, \qquad \Phi_j T_j^- = T_j^+ \Phi_j, \qquad \Phi_j(r_j^-) = r_j^+. \tag{8.13}$$

It is horizontal for the tail and z-connections, preserves the chosen Stokes filtration, is equivariant for the ramified deck action, and commutes with input and bulk derivatives, quotient maps between Artin levels, and the Rees substitution (8.6). At a zero-mode rank change, Φ_j compares the two row-generated nearby-cycle modules, not ambient modules of unequal rank. These maps are compatible over the complete separated inverse system of Artin quotients.

Condition (8.13) concerns only the cyclic module and allows arbitrary connection on the complementary source. A one-object Gamma/window comparison compatible with all structures in Hypothesis 8.6 would imply it: a window mutation is supported on the wall, whereas the skyscraper of a point in the common open is unchanged and Euler-orthogonal to wall-supported objects. Pairing preservation and formal exponential labels alone do not imply (8.13), as Sections 5–7 show.

Lemma 8.7 (Cyclic-row spectral support)**.** *Let T be an endomorphism of a finite dimensional vector space V, and let $r \in V^*$. Put*

$$C_T(r) = \operatorname{span}\{r, rT, rT^2, \ldots\} \subset V^*.$$

For every eigenvalue λ, the restriction of r to the generalized λ-eigenspace of T is nonzero if and only if the λ-primary part of $C_T(r)$ is nonzero.

Proof. Let $e_\lambda(T)$ be the polynomial Bézout projector onto the generalized λ-eigenspace. The relevant primary projection of the row is $re_\lambda(T)$. It belongs to $C_T(r)$ and vanishes exactly when the restriction of r does. □

8.3 The finite cyclic packet

Formal smooth-endpoint quantum Kirwan surjectivity is elementary in this setting. Modulo the positive-energy ideal, $D\kappa_\pm$ is the derivative of the classical Kirwan map and is surjective. Choose a linear section of the reduction. Its arbitrary lift has composite $1 + N$, with N

in the augmentation ideal, and the convergent Neumann series $\sum_{j\geq 0}(-N)^j$ gives a right inverse. Since $D\tau_{Y_\pm,-}$ is an invertible fundamental solution, the maps (8.1) are surjective as well.

The Rees factor in (8.6) is needed to compare formal monodromy. Homogeneity of $D\kappa$ gives

$$\mu_Y A - A\mu_W = \frac{1}{2}A - \Theta_Q A, \tag{8.14}$$

because the acting group has dimension one and $\Theta_Q(Q^d) = c_1^{\mathbf{G}_m}(TW)\cdot d\, Q^d$. After (8.6), the Θ_Q-term becomes $z\partial_z$, so both endpoint maps intertwine the z-connections with the same half-Tate shift. The Euler-product term intertwines because the virtual-dimension identity makes the quantum Kirwan map homogeneous and its derivative a quantum-product morphism. The target bulk point $\kappa(0)$ has positive energy and is formally parallel to the large-radius point; this transport preserves the point row.

Choose finitely many common equivariant inputs whose images under each surjection in (8.1) span the corresponding endpoint module; take the union of two lift sets if necessary. Evaluation on these images embeds the dual endpoint module into one finite coordinate space. After (8.6), this embedding intertwines the dual z-connections up to the same half-Tate shift. Let $T_\pm$ denote the induced half-Tate-normalized formal monodromies. The Krylov spans

$$\operatorname{span}\{\mathfrak{r}_\pm, \mathfrak{r}_\pm T_\pm, \mathfrak{r}_\pm T_\pm^2, \dots\}$$

are the distinguished formal-monodromy subspaces inside the finite dual cyclic modules in that coordinate space. Injectivity here is simply dual to the surjectivity of (8.1).

Proposition 8.1 identifies the marked coefficient functions before either endpoint completion is taken. Theorem 8.4 and Proposition 8.5 supply their holonomic tail systems. Hypothesis 8.6 identifies the finite cyclic Rees z-modules across every threshold, while nonneutral directions are Laurent-finite by Proposition 8.2. Thus the two intrinsic endpoint cyclic row modules are conservative realizations of one continued marked cyclic module: at each Artin level, compose the finitely many threshold maps Φ_j, then pass to the complete inverse system using their reduction compatibility. No continuation of the complementary equivariant quantum source is used.

Theorem 8.8 (Conditional birational invariance of the point-row primary Boolean)**.** *Let Y_- and Y_+ be smooth projective birational complex varieties. Suppose a Włodarczyk completion for this birational map satisfies Hypothesis 8.6 in every neutral affine gauge direction. Then, for every formal-monodromy eigenvalue λ,*

$$\mathfrak{r}_{Y_-}|_{\mathcal{P}_\lambda(Y_-)} \neq 0 \quad \Longleftrightarrow \quad \mathfrak{r}_{Y_+}|_{\mathcal{P}_\lambda(Y_+)} \neq 0. \tag{8.15}$$

Here the packets are realized through the marked cyclic continuation in the hypothesis; no continuation of the complementary source is part of the statement.

Proof. Use the global Włodarczyk completion and its orbit-cylinder class. By the preceding construction, the two endpoint point rows generate the same marked cyclic module under the dual formal-monodromy action. More explicitly, polynomial functional calculus in (8.13) gives

$$\Phi_j\big(r_j^- e_\lambda(T_j^-)\big) = r_j^+ e_\lambda(T_j^+)$$

at every threshold. Hence the relevant primary row vanishes on one side if and only if it vanishes on the other. At a zero-mode threshold this equality is taken in the row-generated nearby-cycle modules; the preserve-and-reflect clause in Hypothesis 8.6 transfers the vanishing statement to the two adjacent tail modules. Iterate through the locally finite threshold system and apply Lemma 8.7 at the endpoints. Because the Artin inverse system is separated, a projected row is nonzero exactly when it survives at some finite level. This proves both implications. □

Under Hypothesis 8.6, Theorem 8.8 is the global replacement for Hypothesis 7.1. Without those inputs, the two-tail example above and the earlier countermodels show why the conclusion cannot be obtained from tailwise holonomicity, localized one-wall receivers, or unmarked Fourier boundary data alone.

9 The cubic endpoint

The conditional transport theorem contains no special feature of cubic threefolds. This section proves the endpoint statement needed for the application. We retain Cai's quantum-connection convention and reconstruct the exact block used below, so the application does not depend on translating a qualitative summary of his result.

9.1 The primitive-sixth block

Let $X \subset \mathbf{P}^4$ be a smooth cubic threefold, let P be its hyperplane class, and put $q = u^2 \neq 0$. In the basis $1, P, P^2, P^3$, Cai writes the small z-equation as

$$z^2 \partial_z S = (K + zG)S, \tag{9.1}$$

where

$$K = 2\begin{pmatrix} 0 & 6q & 0 & 36q^2 \\ 1 & 0 & 15q & 0 \\ 0 & 1 & 0 & 6q \\ 0 & 0 & 1 & 0 \end{pmatrix}, \qquad G = \operatorname{diag}\left(\frac{3}{2}, \frac{1}{2}, -\frac{1}{2}, -\frac{3}{2}\right). \tag{9.2}$$

See [Cai26, Section 3]. Over $\mathbf{Q}(\sqrt{3})(u)$, the eigenvalues of K are $6\sqrt{3}u, -6\sqrt{3}u, 0, 0$, and its zero eigenspace is one rank-two Jordan block. Solving the first off-block Sylvester equation gives, on this block,

$$M_1 = \begin{pmatrix} -19/18 & 0 \\ 0 & 19/18 \end{pmatrix}, \qquad (R_2)_{21} = -16/81, \tag{9.3}$$

when the Jordan link is normalized to one. The indicial polynomial is therefore

$$(\rho + 19/18)(\rho - 1/18) + 16/81 = \rho^2 + \rho + \frac{5}{36}. \tag{9.4}$$

Its roots are $-1/6$ and $-5/6$, hence the formal-monodromy eigenvalues of this block are ζ_6^{-1} and ζ_6. This direct calculation recovers the small-connection part of [Cai26, Proposition 6]. Cai's flatness argument in the same proposition gauges the big connection back to the small one and preserves this rank-two block.

9.2 The point coefficient

The preceding calculation finds the packet but does not say whether the point row detects it. That coefficient follows from one scalar period. The regularized cubic quantum period is

$$F(t) = \sum_{d \geq 0} \frac{(3d)!}{(d!)^5} t^d = {}_2F_3\left(\begin{matrix} 1/3,\ 2/3 \\ 1,\ 1,\ 1 \end{matrix}; 27t\right). \tag{9.5}$$

For the Gamma-framed point class, multiplication by $\widehat{\Gamma}_X$ and $z^{c_1(X)}$ leaves the top class unchanged. Pairing with the unit selects the degree-zero coefficient of the small I-function. Up to one fixed nonzero normalization, its central charge is

$$Z_p(z) = z^{-3/2} F(q/z^2). \tag{9.6}$$

The algebraic part of the large-variable ${}_2F_3$ expansion has simple poles at 1/3 and 2/3; their difference is not integral. The standard Barnes formula [NIS26, Equations 16.11.2 and 16.11.8] gives the leading coefficients

$$C_{1/3} = \frac{\Gamma(1/3)}{\Gamma(2/3)^4}, \qquad C_{2/3} = \frac{\Gamma(-1/3)}{\Gamma(1/3)^4}. \tag{9.7}$$

Both are nonzero. The sector-dependent branch phases of $(-x)^{-1/3}$ and $(-x)^{-2/3}$ are nonzero as well. Substituting $x = 27q/z^2$ in (9.6) gives the powers

$$z^{-3/2}x^{-1/3} \sim z^{-5/6}, \qquad z^{-3/2}x^{-2/3} \sim z^{-1/6}. \tag{9.8}$$

Thus the Gamma point section has a nonzero coefficient in both formal lines found in (9.4). The flat pairing identifies inverse-monodromy lines nondegenerately. Equivalently, the point row is nonzero on both dual lines.

Proposition 9.1 (Cubic endpoint lemma)**.** *For every smooth cubic threefold X,*

$$\mathfrak{r}_X|_{\mathcal{P}_{\zeta_6}(X)} \neq 0, \qquad \mathfrak{r}_X|_{\mathcal{P}_{\zeta_6^{-1}}(X)} \neq 0. \tag{9.9}$$

The packet assertion follows from Cai's connection, while the point-row nonvanishing is the direct Barnes calculation (9.5)–(9.8).

Remark 9.2 (Exact regression)*.* The paper's verification package uses and verifies an explicit basis of $\operatorname{im} K^2 \oplus \ker K^2$, normalizes the nilpotent link, solves the first off-block Sylvester equation, and reconstructs (9.3)–(9.4) at three nonzero rational specializations of the quantum parameter by exact rational arithmetic. It also checks the hypergeometric recurrence behind (9.5) and the projective-space grading calculation below. The proof remains the calculation displayed above.

9.3 The two endpoints

For $\mathbf{P}^m$, write $n = m+1$. At a nonzero quantum parameter $q_{\mathbf{P}}$, multiplication by $c_1(\mathbf{P}^m)$ is a weighted cyclic shift multiplied by n. After adjoining the n-th root of the parameter and rescaling the basis, it is $nq_{\mathbf{P}}^{1/n}$ times the standard cyclic shift. In its Fourier eigenbasis the diagonal of the grading operator is

$$\frac{1}{n}\sum_{k=0}^{m}(m/2 - k) = 0. \tag{9.10}$$

All n rank-one formal factors therefore have formal-monodromy exponent zero. In particular, $\mathbf{P}^{m+3}$ has no primitive-sixth packet.

The quantum connection of a product is the tensor connection. Hence $X \times \mathbf{P}^m$ has $m+1$ copies of each cubic primitive-sixth line. The extended Gamma framing is multiplicative under products [Iri23, Remark 1.2], and the rank row is the tensor product of the two rank rows. Since the rank row of $\mathbf{P}^m$ is nonzero, (9.9) implies

$$\mathfrak{r}_{X\times\mathbf{P}^m}|_{\mathcal{P}_{\zeta_6}(X\times\mathbf{P}^m)} \neq 0, \qquad \mathcal{P}_{\zeta_6}(\mathbf{P}^{m+3}) = 0. \tag{9.11}$$

Proof of Theorem 1.1. If $X \times \mathbf{P}^m$ were rational, it would be birational to $\mathbf{P}^{m+3}$. Theorem 8.8 would identify the two point-row Booleans at ζ_6. Equation (9.11) gives the values one and zero, a contradiction. □

The separation between Theorems 8.8 and 1.1 is deliberate. The conditional transport argument does not use Cai's calculation. Any replacement endpoint with a point-row-detected primary packet gives another application of the same criterion.

10 Scope and source boundaries

The proof combines statements of different kinds. We record their logical roles explicitly.

(1) Theorem 3.1 is an exact identity inside the Gu–Yu–Yu completed formal comparison for $r_+ < r_-$. It concerns only the ambient point coordinate. It is not used to compose the global cobordism.

(2) Theorem 4.1 is an intrinsic point-section identity on one fixed Lee–Lin–Qu–Wang continuation domain. It does not assert full descendant invariance or a Gamma/Fourier–Mukai square for all classes.

(3) Corollary 3.4 requires its named one-wall sectorial realization hypothesis. The formal Gu–Yu–Yu decomposition and the extremal Shen–Shoemaker asymptotics do not by themselves identify the intrinsic large-radius point section in a common receiver.

(4) Propositions 5.1 and 6.1 are exact analytic and algebraic countermodels. They are not asserted to arise from smooth projective quantum connections. They show why the local receivers cannot simply be glued by formal monodromy, pairing, or localized support.

(5) Proposition 8.1 uses the global orbit-cylinder marking. The wall contributions vanish because their attaching nodes land in intermediate fixed loci on which this particular class restricts to zero. No claim is made that arbitrary insertions enjoy the same collapse.

(6) Proposition 8.2 proves the coefficientwise Gamma-ratio and balance identities for individual fixed-graph contributions. Theorem 8.4 supplies the degree-shift identification of the fixed stacks, obstruction theories, virtual classes, and evaluations. Consequently, Proposition 8.5 proves holonomicity and tempered growth for each complete clutching tail, including arbitrary bubble trees.

(7) Hypothesis 8.6 asks for isomorphisms of the finite cyclic Rees z-modules which intertwine formal monodromy and carry the point row across every threshold. Aleshkin–Liu prove a balanced contour theorem for finite-dimensional linear abelian GLSMs; they prove neither this marked compatibility nor its nonlinear virtual fixed-graph analogue. The rational two-tail example shows that separate holonomicity does not determine the threshold maps.

(8) The global proof continues only the finite cyclic module generated by the point row under formal monodromy. It does not assume that the entire equivariant quantum source has finite rank or admits a common continuation. No formal Novikov variable is evaluated at a nonzero complex number.

(9) Cai enters only Proposition 9.1. Equations (9.1)–(9.4) reconstruct the formal block from his displayed matrices, and (9.5)–(9.8) independently compute the point coefficient. The birational conditional transport theorem is unchanged if this endpoint lemma is replaced by any other detected primary packet.

Birational cobordism can have intermediate coarse quotients with finite quotient singularities [Wło00, AKMW02]. The global proof does not form their quantum D-modules. It uses only the smooth projective equivariant completion, the smooth extreme quotients, and the complete fixed loci in the gauged theory. This is why resolving and composing the intermediate walls is unnecessary.

Theorem 8.8 is conditional on Hypothesis 8.6. It concerns the point row on a formal primary packet in the marked cyclic continuation postulated by that hypothesis. It is not a claim that all Stokes matrices, Gamma lattices, or derived categories are birational invariants. The incomplete-Gamma and punctual countermodels rule out those stronger inferences. What survives is exactly one Boolean row, obtained from one globally marked class.

The computational artifact has a similarly narrow role. It recomputes the cubic block and indicial polynomial, the period recurrence, and the projective grading identity. It does not verify virtual localization, the rank-one derived clutching theorem, or marked threshold compatibility. In particular, it cannot turn the conditional birational theorem into an unconditional one.

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References

[AKMW02] Dan Abramovich, Kalle Karu, Kenji Matsuki, and Jarosław Włodarczyk. Torification and factorization of birational maps. *Journal of the American Mathematical Society*, 15:531–572, 2002.

[AL23] Konstantin Aleshkin and Chiu-Chu Melissa Liu. Higgs–coulomb correspondence and wall-crossing in abelian GLSMs, 2023.

[Cai26] Jiaji Cai. The cubic threefold is symplectically irrational, 2026.

[GW15] Eduardo Gonzalez and Chris T. Woodward. A wall-crossing formula for Gromov–Witten invariants under variation of GIT quotient. *Journal of Algebraic Geometry*, 24(1):171–198, 2015.

[GYY25] Zhaoxing Gu, Song Yu, and Tony Yue Yu. Quantum cohomology of variations of GIT quotients and flips, 2025.

[Iri23] Hiroshi Iritani. Gamma classes and quantum cohomology, 2023.

[LLQW16] Yuan-Pin Lee, Hui-Wen Lin, Feng Qu, and Chin-Lung Wang. Invariance of quantum rings under ordinary flops III: A quantum splitting principle. *Cambridge Journal of Mathematics*, 4(3):333–401, 2016.

[NIS26] NIST Digital Library of Mathematical Functions. Generalized hypergeometric functions: Asymptotic expansions. `https://dlmf.nist.gov/16.11`, 2026. Accessed August 2026.

[RSSW21] Thomas Reichelt, Mathias Schulze, Christian Sevenheck, and Uli Walther. Algebraic aspects of hypergeometric differential equations. *Beiträge zur Algebra und Geometrie*, 62(1):137–203, 2021.

[Rud26] Tavis Rudd. Irrationality after one stabilization of universally CH0-trivial cubic threefolds. Preprint, 2026.

[SS25] Yefeng Shen and Mark Shoemaker. Quantum spectrum and gamma structure for standard flips, 2025.

[Wło00] Jarosław Włodarczyk. Birational cobordisms and factorization of birational maps. *Journal of Algebraic Geometry*, 9:425–449, 2000.

[Woo18] Chris T. Woodward. Quantum kirwan morphism and Gromov–Witten invariants of quotients III, 2018.